

My Report

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Table of contents

0.1	Setup and Cleaning for milestone 06	1
0.2	Setup and Linearity	1
0.3	Independence	2
0.3.1	How many tiers do we have	2
0.3.2	Refit and dummy code with conference	3
0.4	Constant Variance	4
0.4.1	Residual spread by tiers	4
0.5	Normality	5
0.6	Identifiability	5
1	Milestone 08	6
1.1	Train and testing split	6
1.2	Transformations	7
1.3	Dummy coding	8
1.4	OLS Model	9
1.5	Interpret	11

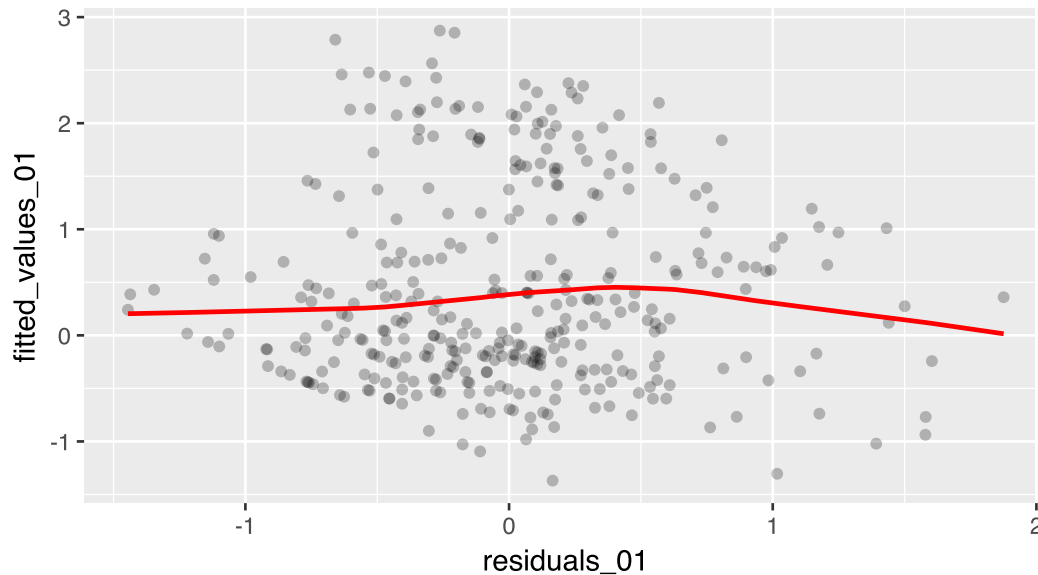
0.1 Setup and Cleaning for milestone 06

0.2 Setup and Linearity

/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/plotnine/stats

Linearity: Residuals vs. Fitted Values

Look for a roughly horizontal line



The line pretty good and is pretty horizontal. There is a little wobble where there are less teams. We also already did a log-transform to NIL so the linearity assumption looks satisfied

0.3 Independence

0.3.1 How many tiers do we have

```
n_tiers
u32
1
```

conference	teams	mean_residual
str	u32	f64
"Patriot"	8	-0.761513
"Southland"	10	-0.609862
"MissouriValley"	10	-0.478225
"MAAC"	13	-0.413068
"MEAC"	8	0.610889
"BigEast"	11	0.623245
"Atlantic10"	14	0.67061
"WestCoast"	9	0.892118

So every conference has exactly one tier, so tier is just a coarse grouping of conferences. If the residuals were independent, each conference's average residual would be near 0. They aren't. so the

independence assumption is not satisfied for right now. The remedy is to add a predictor for the dependence. We replace tier with conference dummies, because tier would be perfectly redundant with them.

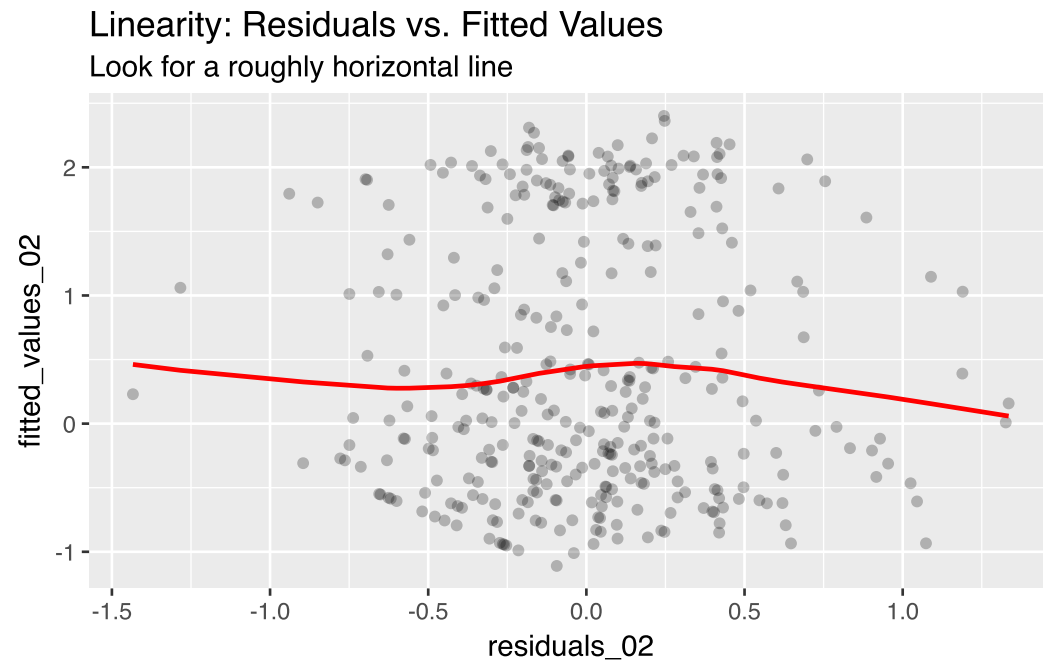
0.3.2 Refit and dummy code with conference

```
(np.float64(0.7389089340294595),  
 np.float64(0.8583042414351332),  
 np.float64(0.8441346655786465),  
 np.float64(0.05742840447356816),  
 np.float64(0.025350754162115217))
```

Adding conferences raised our r squared

0.3.2.1 Check linearity again rq

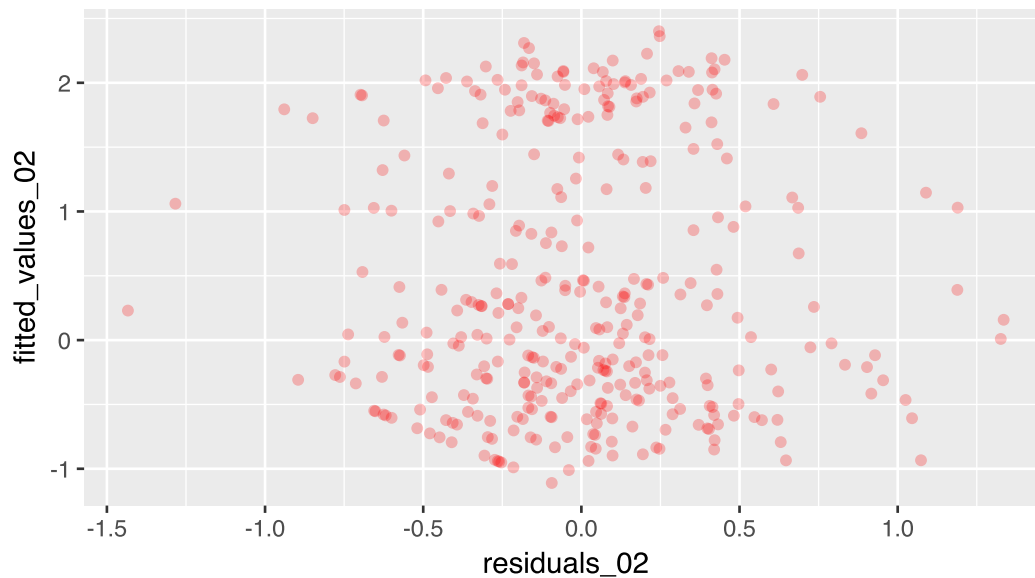
```
/Library/Frameworks/Python.framework/Versions/3.14/lib/python3.14/site-packages/plotnine/stats,
```



0.4 Constant Variance

Constant Variance: Residuals vs. Fitted Values

Look for a funnel shape in the residuals

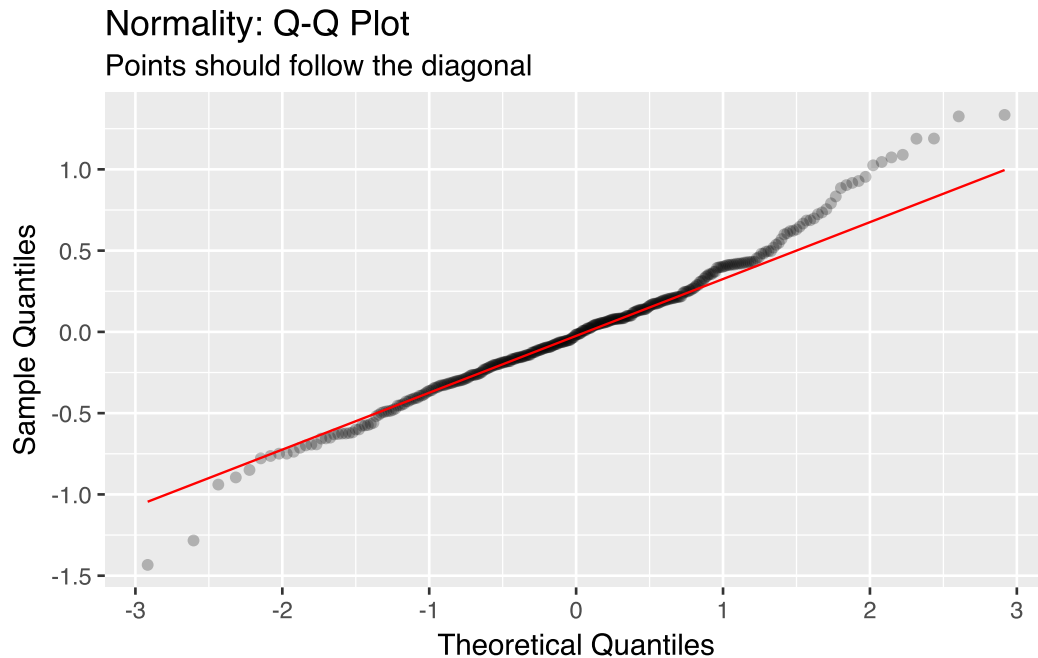


0.4.1 Residual spread by tiers

tier	teams	sd_residual
str	u32	f64
"Ivy"	8	0.165528
"Power4"	68	0.28777
"G5"	53	0.412269
"LowerTier"	100	0.434507
"FCS"	68	0.458273
"Basketball"	56	0.464532

I don't see much of a funnel in this. Spread is pretty similar across the fitted values. The residual standard deviation is between 0.29 and 0.47 for every tier except Ivy, which has 0.17 but only 8 teams. The constant variance assumption is probably satisfied.

0.5 Normality



0.5.0.1 Teams with the largest positive residuals

team	tier	bpr	nil_m	residuals_02
str	str	f64	f64	f64
"Drake"	"Basketball"	3.68482	4.45	1.334967
"Bradley"	"Basketball"	-2.1846	3.8	1.325859
"Memphis"	"G5"	5.775717	9.2	1.189559
"Massachusetts"	"G5"	0.062609	4.85	1.188564
"Saint Mary's"	"LowerTier"	4.977795	9.35	1.089467
"New Orleans"	"LowerTier"	-8.632951	1.15	1.073655

Most of the line follows closely. The top of it is a little skewed since their NIL is so high than what you would expect. The lower tail is slightly short, partly because 18 teams sit at Sideline's \$0.3M floor. The normality assumption is approximately satisfied, with a mild right-tail departure. These teams are worth investigating rather than removing.

0.6 Identifiability

predictor	VIF
str	f64
"bpr"	4.477024
"conference_SWAC"	2.665913

predictor str	VIF f64
"conference_MAAC"	2.418224
"conference_SunBelt"	2.293987
"conference_MEAC"	2.184331
...	...
"conference_IvyLeague"	1.689709
"conference_WestCoast"	1.648471
"conference_BigEast"	1.562276
"conference_Pac12"	1.51648
"conference_MountainWest"	1.364279

The largest VIF is bpr at 4.48 and then its SWAC at 2.67. The mean is 2.02. None is anywhere near 10, so the identifiability assumption is satisfied.

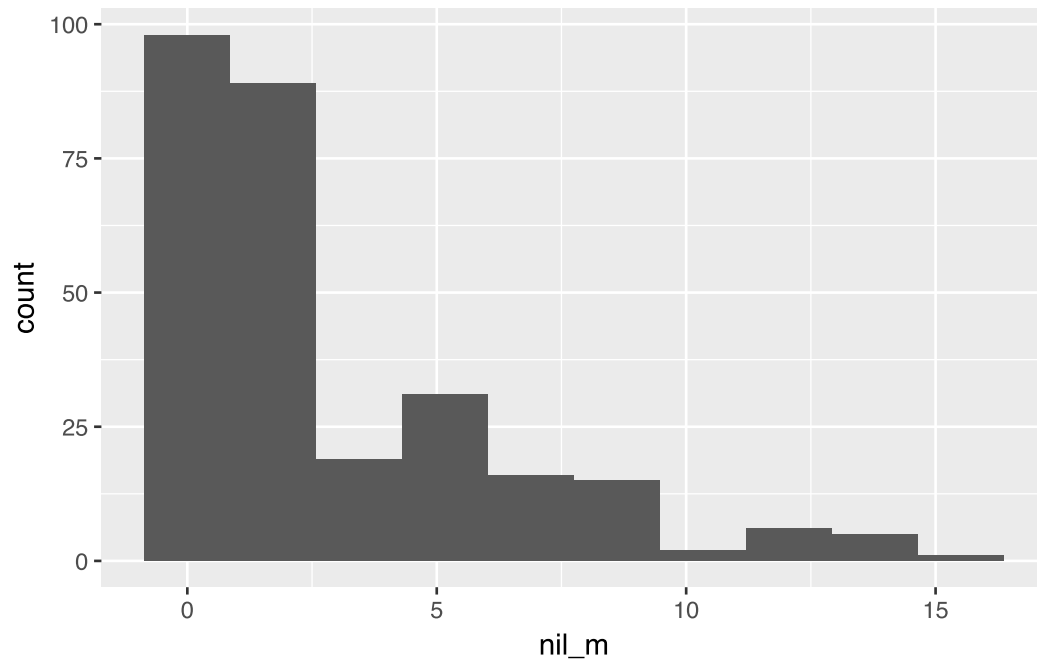
1 Milestone 08

We gotta continue here because our unique function returns rows in a different order every time so our numbers won't match if we do a new file

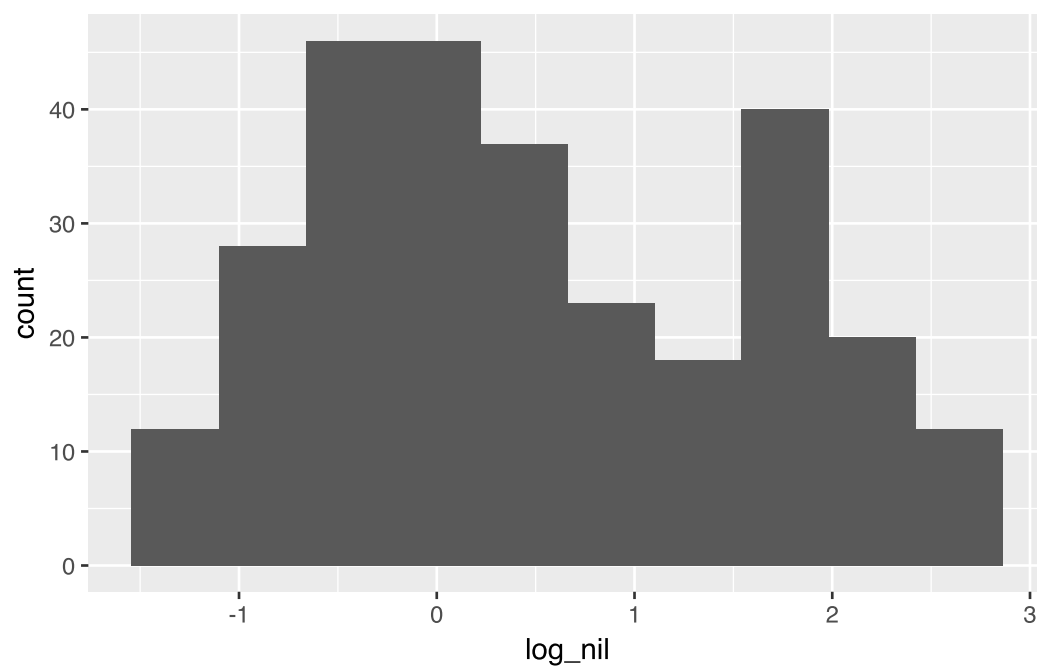
1.1 Train and testing split

282 for training and 71 for testing

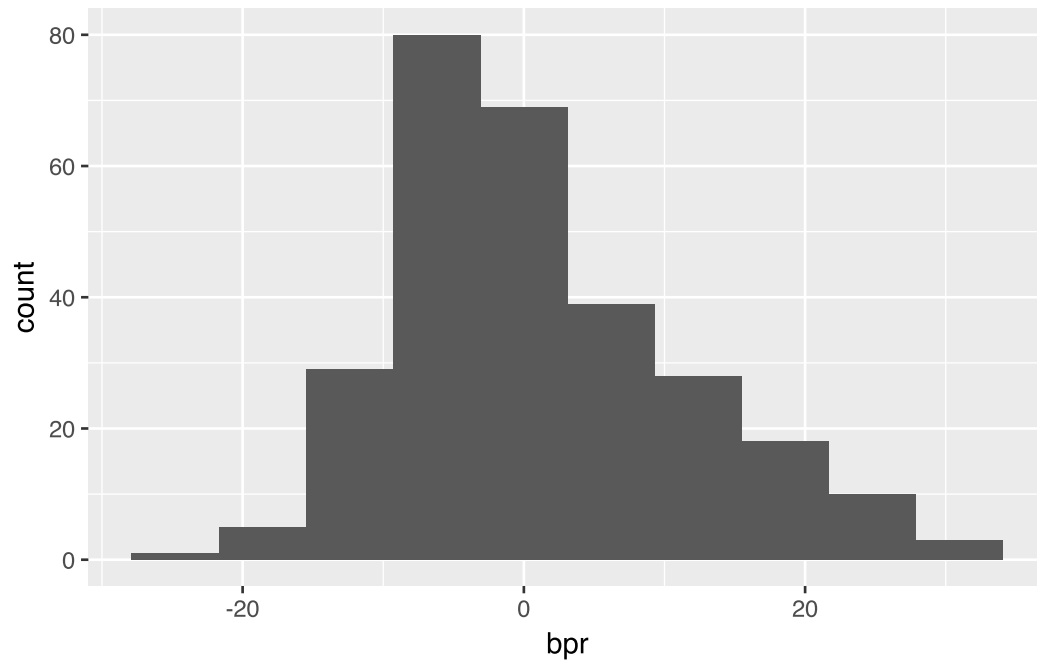
1.2 Transformations



So as we knew, NIL is right skewed. Most of the teams sit under \$3M, with a long tail to \$16M. So we give it a log transform since no value is under 0



Skew is gone. The two humps are the power conference teams and everyone else.



BPR is roughly symmetric, so we don't gotta transform anything

1.3 Dummy coding

Conference has 32 levels, so we get 31 dummy variables. Big Ten is the baseline. It is tied with the ACC as the largest conference at 18 teams and sits at the top of the market, so every other conference estimate reads as a gap from the top. Tier is not included, because it is perfectly nested in conference

1.4 OLS Model

Dep. Variable:	log_nil	R-squared:	0.856
Model:	OLS	Adj. R-squared:	0.838
Method:	Least Squares	F-statistic:	46.34
Date:	Fri, 02 Oct 2026	Prob (F-statistic):	6.93e-87
Time:	15:48:11	Log-Likelihood:	-147.13
No. Observations:	282	AIC:	360.3
Df Residuals:	249	BIC:	480.4
Df Model:	32		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.4183	0.143	9.914	0.000	1.137	1.700
bpr	0.0283	0.005	5.360	0.000	0.018	0.039
conference__ACC	0.1637	0.165	0.994	0.321	-0.161	0.488
conference__ASUN	-1.6011	0.216	-7.427	0.000	-2.026	-1.177
conference__AmericaEast	-1.8503	0.230	-8.051	0.000	-2.303	-1.398
conference__American	-0.5550	0.213	-2.608	0.010	-0.974	-0.136
conference__Atlantic10	-0.0907	0.179	-0.507	0.613	-0.443	0.262
conference__Big12	0.0534	0.167	0.319	0.750	-0.276	0.383
conference__BigEast	0.1582	0.188	0.840	0.402	-0.213	0.529
conference__BigSky	-1.7216	0.224	-7.676	0.000	-2.163	-1.280
conference__BigSouth	-1.6357	0.226	-7.238	0.000	-2.081	-1.191
conference__BigWest	-1.1801	0.204	-5.781	0.000	-1.582	-0.778
conference__CAA	-1.4032	0.207	-6.769	0.000	-1.812	-0.995
conference__ConferenceUSA	-1.0441	0.203	-5.141	0.000	-1.444	-0.644
conference__Horizon	-1.4843	0.217	-6.848	0.000	-1.911	-1.057
conference__IvyLeague	-2.2643	0.222	-10.213	0.000	-2.701	-1.828
conference__MAAC	-1.4737	0.210	-7.022	0.000	-1.887	-1.060
conference__MAC	-1.0250	0.196	-5.219	0.000	-1.412	-0.638
conference__MEAC	-0.6771	0.300	-2.256	0.025	-1.268	-0.086
conference__MissouriValley	-1.2328	0.211	-5.844	0.000	-1.648	-0.817
conference__MountainWest	-0.9084	0.219	-4.156	0.000	-1.339	-0.478
conference__NEC	-1.8057	0.267	-6.754	0.000	-2.332	-1.279
conference__OhioValley	-1.8332	0.256	-7.149	0.000	-2.338	-1.328
conference__Pac12	-0.4677	0.192	-2.442	0.015	-0.845	-0.090
conference__Patriot	-1.8578	0.297	-6.263	0.000	-2.442	-1.274
conference__SEC	0.0985	0.163	0.605	0.546	-0.222	0.419
conference__SWAC	-1.6606	0.239	-6.939	0.000	-2.132	-1.189
conference__SoCon	-1.3438	0.247	-5.448	0.000	-1.830	-0.858
conference__Southland	-2.0902	0.215	-9.710	0.000	-2.514	-1.666
conference__SummitLeague	-1.3459	0.236	-5.715	0.000	-1.810	-0.882
conference__SunBelt	-0.8120	0.210	-3.863	0.000	-1.226	-0.398
conference__WAC	-1.8582	0.230	-8.074	0.000	-2.311	-1.405
conference__WestCoast	-0.4035	0.200	-2.015	0.045	-0.798	-0.009

Omnibus:	11.346	Durbin-Watson:	2.012
Prob(Omnibus):	0.003	Jarque-Bera (JB):	13.001
Skew:	0.380	Prob(JB):	0.00150
Kurtosis:	3.728	Cond. No.	344.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

shape: (32, 3)

term	estimate	pct_change
---	---	---
str	f64	f64
conference_ACC	0.16367	17.782614
conference_BigEast	0.158174	17.13702
conference_SEC	0.098462	10.347238
conference_Big12	0.053401	5.485272
bpr	0.028263	2.866655
conference_Atlantic10	-0.090747	-8.675171
conference_WestCoast	-0.403466	-33.199905
conference_Pac12	-0.467733	-37.357954
conference_American	-0.555025	-42.594224
conference_MEAC	-0.677056	-49.188946
conference_SunBelt	-0.812032	-55.604478
conference_MountainWest	-0.908436	-59.68459
conference_MAC	-1.02499	-64.119982
conference_ConferenceUSA	-1.044087	-64.798709
conference_BigWest	-1.180111	-69.275547
conference_MissouriValley	-1.232822	-70.853121
conference_SoCon	-1.34377	-73.913958
conference_SummitLeague	-1.345922	-73.970039
conference_CAA	-1.40323	-75.419825
conference_MAAC	-1.473665	-77.091566
conference_Horizon	-1.484252	-77.332815
conference_ASUN	-1.601118	-79.832905
conference_BigSouth	-1.635717	-80.518733
conference_SWAC	-1.660627	-80.998013
conference_BigSky	-1.72161	-82.122188
conference_NEC	-1.805664	-83.563473
conference_OhioValley	-1.833247	-84.010646
conference_AmericaEast	-1.850334	-84.281528
conference_Patriot	-1.857802	-84.398487
conference_WAC	-1.858164	-84.404134
conference_Southland	-2.090173	-87.633424
conference_IvyLeague	-2.26434	-89.610137

1.5 Interpret

So the model explains most of the variation in log NIL with r squared of 0.856 and an adjusted r squared of .838. The residual standard deviation is 0.434 on the log scale.

- Intercept: The expected log NIL of a Big Ten team with a BPR of 0 is 1.418, or $\exp(1.418)$ \$4.13M. This is a baseline, not a real team. Big Ten BPR ranges from 5.2 to 29.8, so none sits at 0.
 - bpr: Holding all else true, Within the same conference, one more BPR point is associated with $(\exp(0.0283) - 1) \times 100 \approx +2.9\%$ team NIL, holding conference fixed. Ten points, about one standard deviation of BPR, is $\exp(0.283) - 1 \approx +32.7\%$.
 - Conference: ach estimate is the expected percent difference in NIL relative to a Big Ten team with the same BPR.
- The ACC, Big East, SEC and Big 12 are all close to the Big Ten. Their estimates are small compared to the other conferences.
 - Group of 5: The A10, West Coast, Pac-12, American and Sun Belt pay a lot less with the same BPR
 - Bottom Market: All of these conferences have low BPR and low NIL